POKERBOTS

MIT 6.9630

Massachusetts Institute of Technology
6.9630 Pokerbots Competition

IAP 2026
Lectures MTWR 12:00-1:30 in 6-120
Recitation/Office Hours 2-4 in 26-168
Zoom Link: pkr.bot/zoom

Staff

pokerbots@mit.edu

Sri Saraf, *Co-President*	srisar@mit.edu
Bobby Costin, Co-*President*	bcostin@mit.edu
Sejal Rathi, *Vice President*	sejalr@mit.edu
Jacob David, *Head of Curriculum*	jcob@mit.edu
Jacob Jobraeel, *Innovation Chair*	jobraeel@mit.edu
Paco Gomez-Paz, *Engine Lead*	pjgomez@mit.edu
Sam Zhou, *Tech Lead*	samtzhou@mit.edu
Steve Zhang, *Server Lead*	stevez@mit.edu
Landon Sims, *Engine Developer*	lzsims@mit.edu
Reese Long, *Server Developer*	reesel@mit.edu
Alan Duan, *Treasurer*	alanduan@mit.edu
Michelle Gao, *Sponsorship Director*	mgao7@mit.edu

Links

Homepage:	pokerbots.org
Resources:	pkr.bot/resources
Piazza:	pkr.bot/piazza
Scrimmage Server:	pkr.bot/scrimmage
Lecture Recordings	pkr.bot/panopto
Canvas page:	pkr.bot/canvas
Find Teammates:	pkr.bot/teammates
Instagram:	pkr.bot/instagram
Github:	pkr.bot/github

Introduction

Pokerbots is an annual month-long programming competition in which students form teams of 1-4 members to create an autonomous poker-playing algorithm (a "pokerbot"). Over the four weeks of IAP, the open-ended project in pokerbot design builds skills in computer science, game theory, mathematics, and machine learning.

Students will learn how to think about developing their pokerbots by attending ~twelve lectures and will have the chance to test their skills in competition against their peers. In class, students will learn about topics ranging from poker strategy to reinforcement learning, and their applications to building a successful pokerbot. A scrimmage server will be run throughout the month allowing teams to challenge each other at any time. There will be a mini tournament at the end of each of the first three weeks. In the fourth week, teams may continue to work on their pokerbots separately. During the final tournament, teams' finished pokerbots are put to the test in a tournament for exciting prizes.

Poker is a complex yet highly accessible challenge: it's easy to learn, but difficult to be competitive in. Building a pokerbot is similar, as anyone can make an elementary bot, but building a competitive bot requires a deep understanding of the strategies adopted by other teams and how to contest them. At the heart of the Pokerbots competition is the challenge of applying the algorithmic and strategic thinking taught in theoretical courses.

Evaluation

6.9630 is graded on a pass/fail basis. To receive a passing grade, students are expected to fulfill both of two requirements: participation on the scrimmage server, and completion of a final strategy report. All teams who meet the below requirements in good faith will receive a passing grade.

Participation

Pokerbots is designed to encourage exploration. Students who put more effort into trying new ideas to design a winning pokerbot will have greater learning opportunities and a better chance to win prizes. To earn credit, students are expected to put sustained effort into developing their pokerbots. This will be judged by submissions each week during which each team's pokerbot will be evaluated for improved performance relative to their previous week's submission and against a bot that implements a random strategy.

Make-up

If a team fails to submit their pokerbot or improve their pokerbot from one week to the next, that team is expected to submit a half-page double-spaced make-up report describing the attempted improvements for the week and thoughts on why they were unsuccessful. This is intended to avoid penalizing students who explore avenues that do not end up resulting in successful strategies. Make-up reports (if necessary) are due the Tuesday following each mini tournament. Reports should list all team members and be emailed to **pokerbots@mit.edu**

Final Report

To earn credit in this course, each team is required to submit a final strategy report outlining their work up to and including their final pokerbot:

- Submission required from ≥1 member, 3-5 pages, double spaced, PDF format
- Top of first page, must list: Team name and each member's full name and kerb (or email if no kerb)
- **Due Tue 1/27 11:59PM EST** on Canvas: **pkr.bot/canvas**

These reports are meant to catalog the effectiveness of techniques explored throughout the class, as well as describe the distribution of work. In particular, the report should highlight the strategic and/or architectural insights that went into your team's pokerbot, and primary takeaways from the month. Diagrams are optional, but *highly* recommended.

Structure

Over the course of IAP, students have five main touchpoints with the course: class sessions, Piazza, the scrimmage server, Github, and the final event.

Class

There will be twelve lectures in person every Monday through Thursday (6-120, 12-1:30PM), each which will be applicable to developing a successful pokerbot. These lectures will be recorded. Recitation and Office Hours will be held after each lecture (26-168, 2-4PM), as well as on most non-lecture days (info will be posted on Piazza). Each lecture will have provided lunch and one or more giveaways with nice prizes! All of these sessions are available virtually at pkr.bot/zoom, which will be updated to always direct to the correct zoom link.

Piazza

Piazza is an online forum we will be using to answer questions and make announcements. It's important to check Piazza regularly for updates and changes we may make. You can post (publicly or anonymously to everyone or the instructors) and a member of the Pokerbots team will respond as soon as possible (last year, the average response time was less than 5 minutes!). You are also encouraged to answer other students' questions, and we will be rewarding students who contribute the most in this manner over the month. Note that an MIT email address is required to join - if you have issues, please contact the staff.

Scrimmage Server

The scrimmage server is how you will gauge the performance of your pokerbot relative to reference bots and other teams. On the scrimmage server, you can challenge your opponents and keep track of useful bot statistics. We have a Playground feature where you can manually interact with your own or another's pokerbot to get a feel for how it operates in practice and try to beat your bot at the variant! Furthermore, all mini-tournament submissions are done by uploading and selecting your latest bot on the scrimmage server. We'll be giving out prizes based on your performance on the server as well!

Github

Throughout this course we will be posting materials, and additional resources to Github. This is where the Engine repository can be found, as well as any sample code from lecture. We will be uploading slides and a recording after every lecture as well.

Final Event

Pokerbots culminates in a virtual final event on January 30st, 2025. This is where we will announce the winners of the final pokerbots tournament, as well as give out all of our biggest prizes. There will be fun and games, as well as a chance to meet and network with our sponsors directly! More details about the final tournament and event will be posted on the course Piazza as the date approaches.

Other Events

We will be hosting a variety of other events such as afternoon study breaks, poker socials, hackathons, and sponsor networking events. These events are more informal and spontaneous, and information or details would be on Piazza as they approach.

Timeline

Week 1	
Mon 1/5	Lecture 1: Intro to Pokerbots Recitation: Environment Setup
Tue 1/6	Lecture 2: Probability and Statistics Recitation: Practice!
Wed 1/7	Lecture 3: Poker Strategy Recitation: Poker Social!
Thu 1/8	Lecture 4: Machine Learning Recitation: TBD
Fri 1/9	Week 1 Bot due 11:59pm
Sat 1/10	**Mini Tournament #1**
Week 2	
Mon 1/12	TBD
Tue 1/13	TBD
Wed 1/14	TBD
Thu 1/15	No Class: Hackathon
Fri 1/16	Week 2 Bot due 11:59pm
Sat 1/17	**Mini Tournament #2**
Week 3	
Mon 1/19	No Class: MLK Day
Tue 1/20	TBD
Wed 1/21	TBD
Thu 1/22	TBD
Fri 1/23	Week 3 Bot due 11:59pm
Sat 1/24	**Mini Tournament #3**
Week 4	
Mon 1/26	TBD
Tue 1/27	No Class: Final Report/Bot Strategy Report due 11:59pm
Wed 1/28	TBD Final Bot due 11:59pm
Fri 1/30	**Pokerbots Final Event**
Sat 1/31	Poker Tournament (with Prize Pool)

Prizes

The prize pool for Pokerbots 2025 is over **$50,000.** Here's the breakdown:

Final Tournament Prizes	
First place	$10,000
Second place	$6,500
Third place	$3,500
Fourth place	$2,000
Fifth place	$1,000
First place in language (Python, Java, or C++)	$500 x 3
Second place in language (Python, Java, or C++)	$250 x 3
Third place in language (Python, Java, or C++)	$125 x 3
Best freshman-majority (>51%) team	$2,000

Scrimmage Server Prizes	
Weekly tournament winner	$1,000 x 3
Weekly tournament biggest upset	$500 x 3
Weekly tournament most improved	$750 x 2
Most time at the top of the scrimmage server	$1,000

Miscellaneous Prizes	
Most helpful Piazza students	$250 x 3
Surprise prizes, lightning tournaments, raffles	$15,000

We'll be holding raffles during each of the classes and events for great tech prizes!

Sponsors

Platinum

Hudson River Trading (HRT) is a leading quantitative trading firm at the forefront of technical innovation in global financial markets. Every day, we bring together the world's sharpest minds to collaboratively solve challenging problems and design the technology that will drive the future of trading. Leveraging one of the world's most sophisticated computing environments for quantitative research and development, we trade across multiple time horizons and asset classes on more than 200 markets worldwide from our global offices. We are a leading voice for fair and transparent markets everywhere and dedicated to creating a better trading landscape for all.

Jump Trading is a global algorithmic trading firm that combines sophisticated quantitative research, cutting-edge technology, and an entrepreneurial culture. Founded in 1999, Jump has 1,700+ employees across offices in Chicago, New York, London, Amsterdam, Paris, Singapore, Shanghai, Mumbai, Sydney, and Hong Kong.

Jump's culture of continuous learning and innovation has attracted some of the most brilliant minds from around tech, trading, and research. We've built one of the largest supercomputers in the world to allow our team to test their most sophisticated ideas, and we continuously develop custom software and hardware to maximize the impact of our strategies.

Gold

Jane Street is a quantitative trading firm with offices worldwide. We hire smart, humble people who love to solve problems, build systems, and test theories. Will our next great idea come from you?

Gold (continued)

Citadel is a multi-strategy alternative investment manager. For more than three decades, we've captured undiscovered market opportunities by empowering extraordinary people to pursue their best and boldest ideas. Each day at Citadel, we reimagine and refine our strategies, models and technology in pursuit of superior long-term performance.

Citadel Securities is the next-generation capital markets firm. We create efficient ways for buyers and sellers to come together; we innovative ways to price and absorb risk; and we develop new technologies to transform markets. Integrating deep financial, mathematical and engineering expertise, Citadel Securities delivers critical liquidity to central banks, financial institutions, government agencies, corporations, insurers and sovereign wealth funds.

Both Citadel and Citadel Securities operate in global financial centers including Miami, Chicago, Hong Kong, London and New York.

DRW is a diversified trading firm with over three decades of experience bringing sophisticated technology and exceptional people together to operate in markets around the world. We value autonomy and the ability to quickly pivot to capture opportunities, so we trade using our own capital and at our own risk.

Headquartered in Chicago with offices across the U.S., Canada, Europe, and Asia, we participate in major global markets across multiple asset classes including Fixed Income, ETFs, Equities, FX, Commodities, and Energy. We've also expanded our expertise into three non-traditional strategies: real estate, venture capital, and cryptoassets.

We operate with respect, curiosity, and open minds. The people who thrive here share our belief that it's not just what we do that matters—it's how we do it. DRW is a place of high expectations, integrity, innovation, and a willingness to challenge consensus.

Gold (continued)

Sphinx AI (www.sphinx.ai/) builds frontier AI agents for data science. Our agents combine statistical algorithms, multimodal understanding, and business logic to accelerate the discovery of value from data. Sphinx's copilot integrates seamlessly into most data science IDEs (Jupyter/VSCode), compute layers, databases and data lakes. We augment every step of the data science process from data discovery, to exploratory analysis, to modeling, to impactful data-driven storytelling.

Balyasny Asset Management (BAM) is a diversified global investment firm founded in 2001 by Dmitry Balyasny, Scott Schroeder, and Taylor O'Malley. The firm's investment teams span five strategies, including Equities Long/Short, Fixed Income & Macro, Commodities, Multi-Asset Arbitrage, and Systematic. Balyasny's mission is to deliver to its investors absolute, uncorrelated returns in all market environments.

We're a global investment and technology development firm headquartered in New York City with more than 3,000 people around the globe. Since our founding in 1988, we've earned a worldwide reputation for innovation, rigorous risk management, cutting-edge technology, and the quality of our people. To learn more about our culture, impact, research, and programs, visit our website and our YouTube channel.

Gold (continued)

Susquehanna is a global quantitative trading firm powered by scientific rigor, curiosity, and innovation. Our culture is intellectually driven and highly collaborative, bringing together researchers, engineers, and traders to design and deploy impactful strategies in our systematic trading environment. To meet the unique challenges of global markets, Susquehanna applies machine learning and advanced quantitative research to vast datasets in order to uncover actionable insights and build effective strategies. By uniting deep market expertise with cutting-edge technology, we excel in solving complex problems and pushing boundaries together.

Silver

Five Rings is a proprietary trading firm founded with a vision of combining strategy, innovation and technology to succeed in today’s global markets. With offices in New York, London, Amsterdam, and Boca Raton, Five Rings trades in various domestic and international markets, both established and esoteric.